

FACE: ArcFace White-box Geometric Identity Optimization

Frozen iResNet-100 identity-distance results with downstream InstructPix2Pix evaluation

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FACE optimizes $Z = 1 - \text{cosine_similarity}$ between original and perturbed full-image ArcFace iResNet-100 embeddings. The optimized loss is exactly $\text{loss} = -Z$. ArcFace weights are frozen; only geometry parameters are optimized.

No landmarks, face detection, alignment, visual counter-loss, VAE objective, or UNet objective is used. Prompts are retained only for downstream InstructPix2Pix evaluation.

Run matrix and final values

face	prompt	final Z	best Z	cosine sim	score %	SSIM	PSNR	edit SSIM	max disp
face_002	add black sunglasses	0.0592	0.0593	0.9408	94.081	0.9737	24.621	0.6466	6.3730
face_002	add headphones	0.0680	0.0681	0.9320	93.198	0.9733	24.551	0.6269	7.2244
face_005	add black sunglasses	0.1030	0.1032	0.8970	89.704	0.9619	22.976	0.6888	18.171
face_005	add headphones	0.0990	0.0991	0.9010	90.102	0.9613	22.908	0.6790	11.920

Case image strips

face_002 / add black sunglasses



face_002 / add headphones



face_005 / add black sunglasses

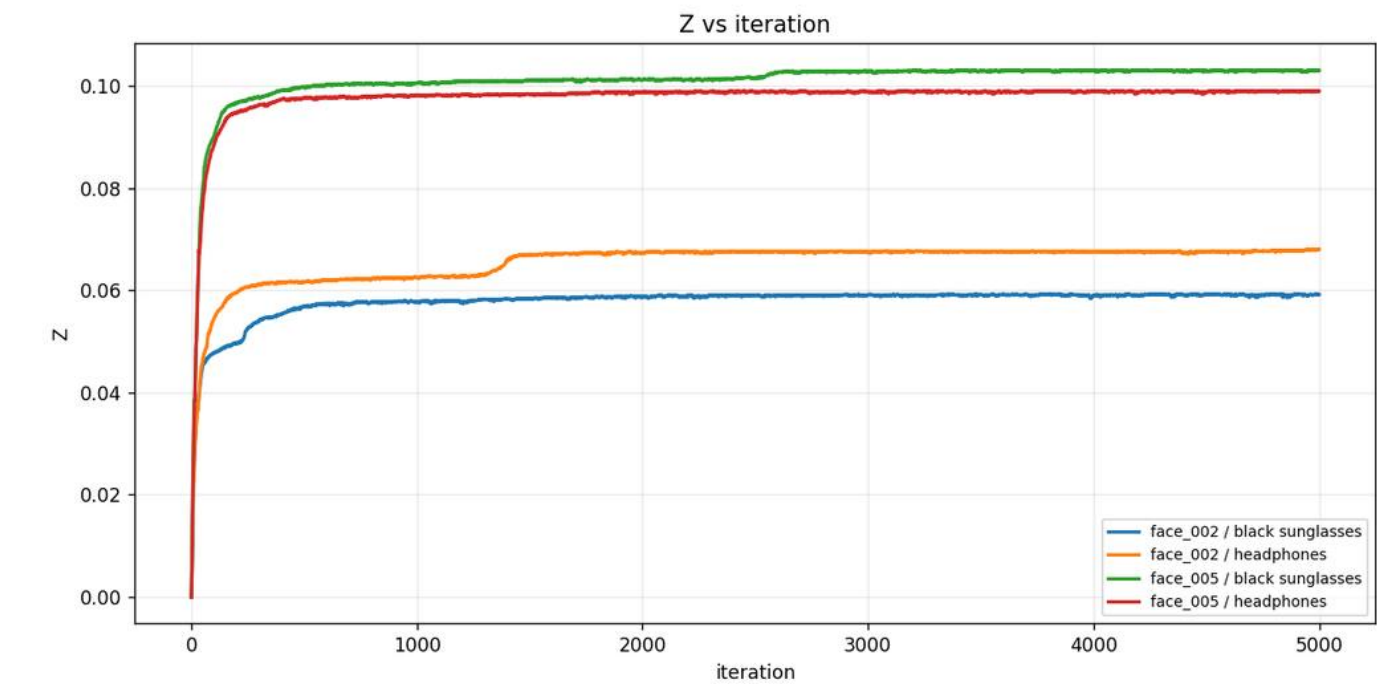


face_005 / add headphones

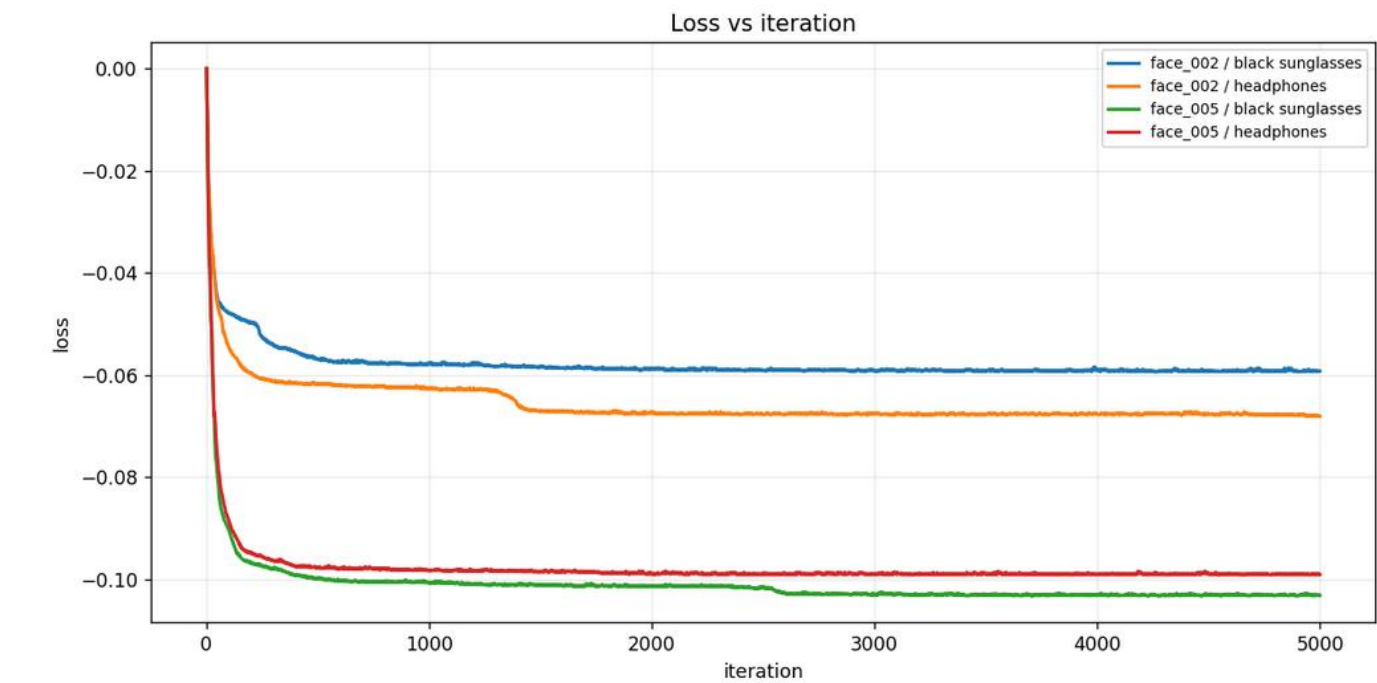


Graphs

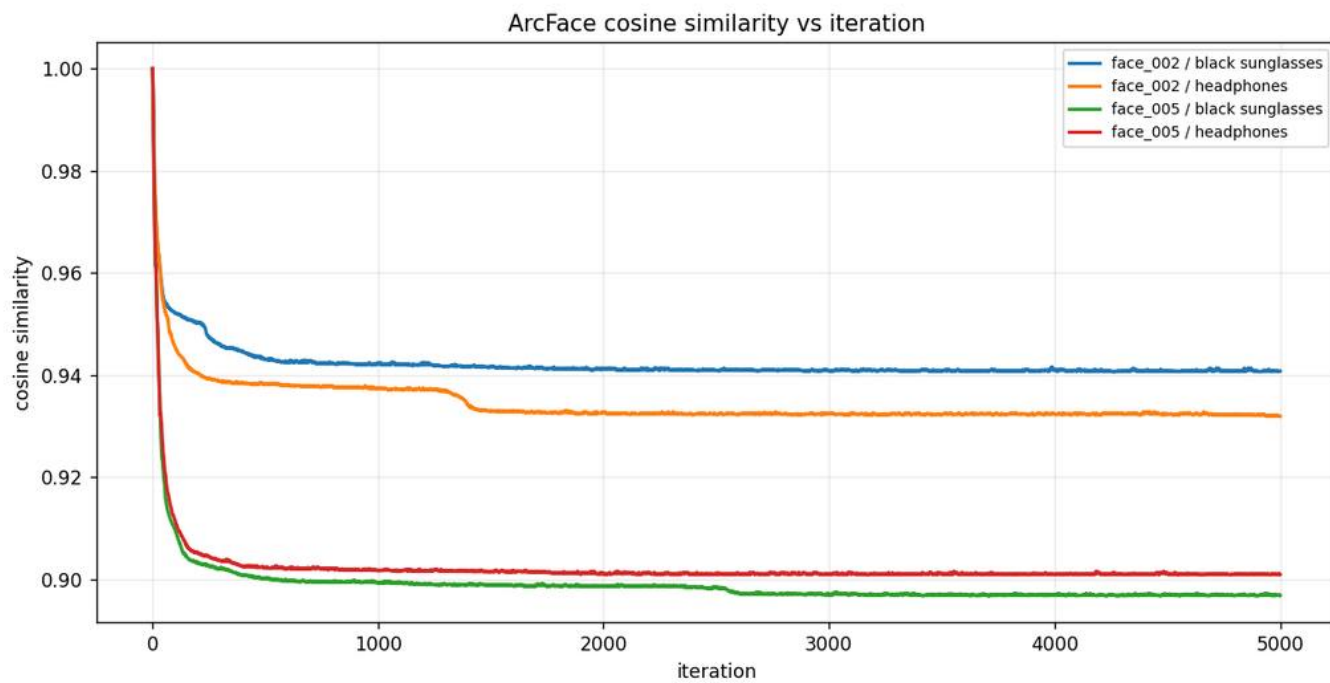
Z vs iteration



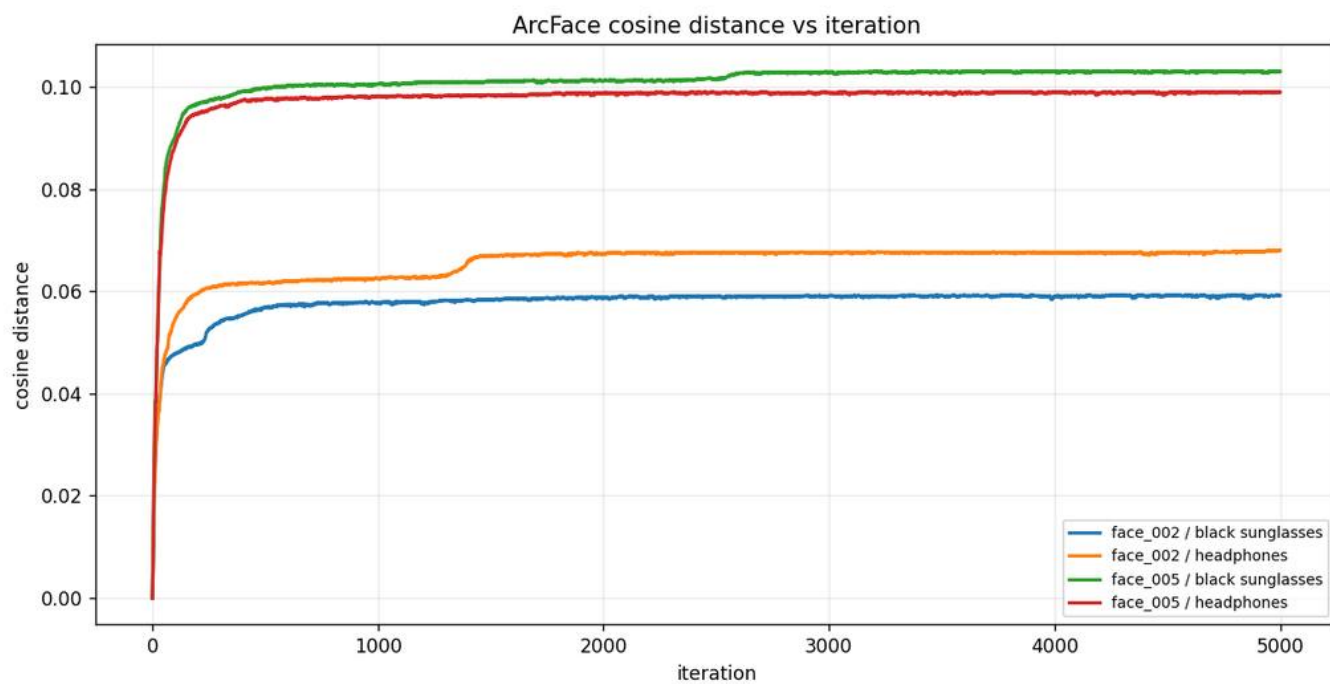
Loss vs iteration



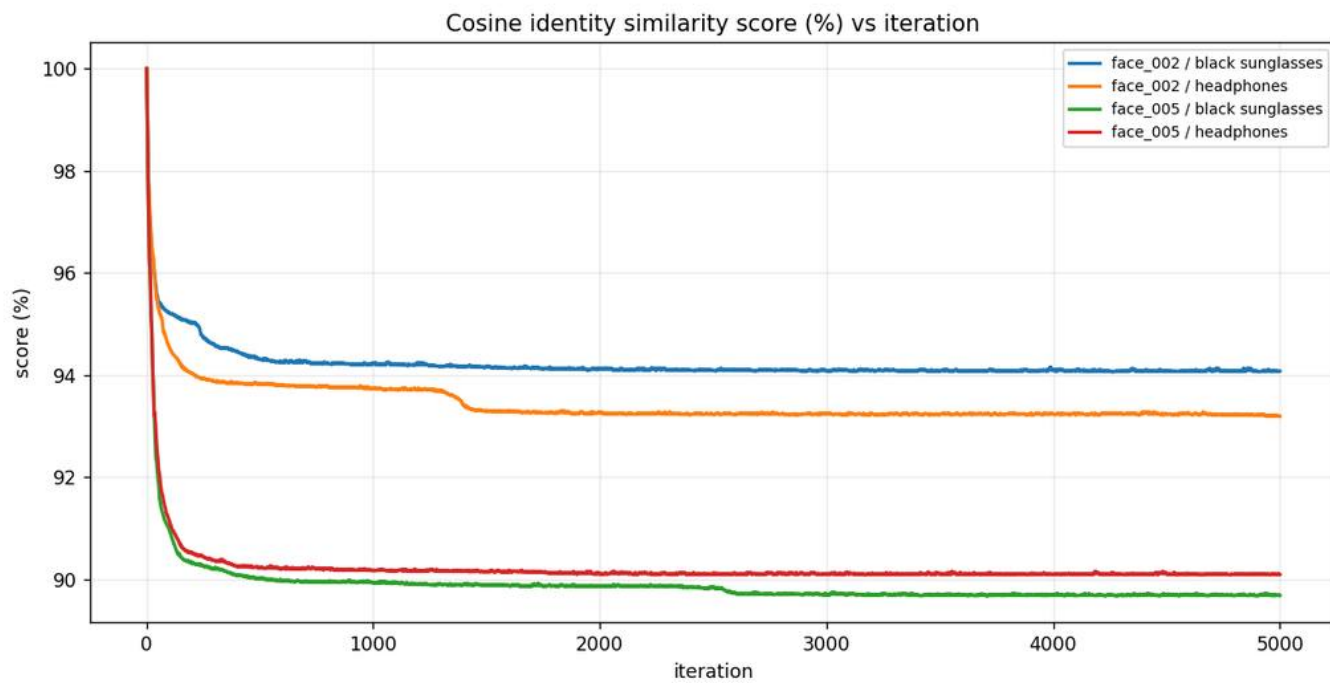
ArcFace cosine similarity vs iteration



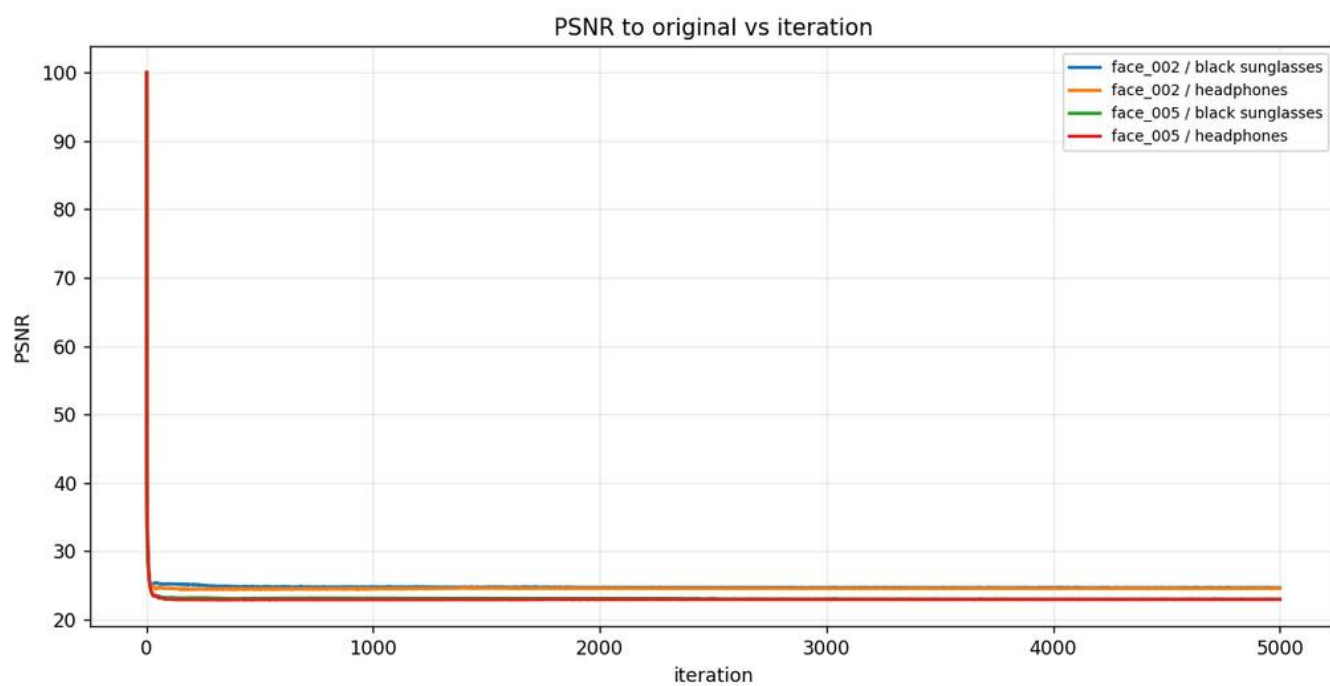
ArcFace cosine distance vs iteration



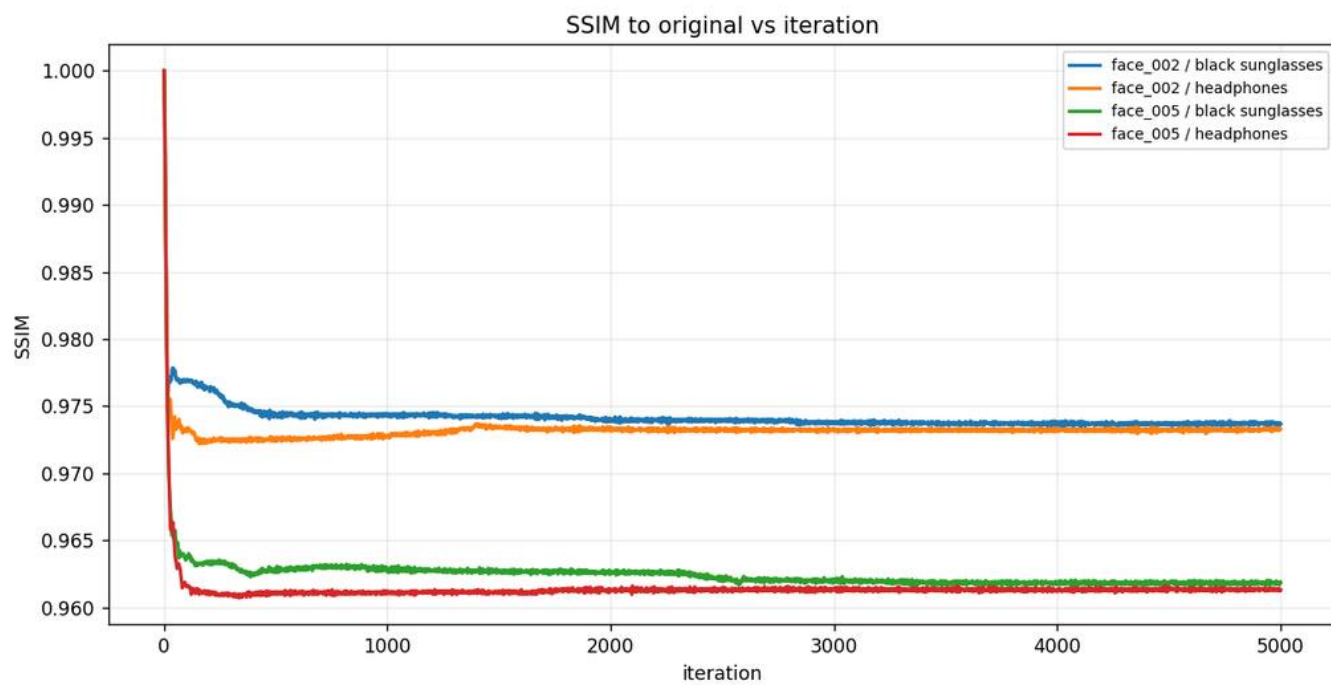
Cosine identity similarity score (%) vs iteration



PSNR to original vs iteration



SSIM to original vs iteration



Notes

Completed runs collected: 4.

Run root: outputs/arcface_identity/20260703_103418_arcface_identity_all_sequential

Missing artifacts recorded: 0.